

Subject: ELC 314  
Electronic Measurements

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Level: 2<sup>nd</sup>

Spec: Elec. Eng. & Comm. Tech.  
Comp. Eng. & Info. Tech.

## Assignment 2

**Write the answer on file word and upload on Moodle website (word and pdf files)**

### Question 1: (1 point)

- 1) A digital frequency meter uses a time base consisting of a 1 MHz clock generator frequency-divided by six decade counters. Determine the meter indication (a) when the input frequency is 5 kHz and the time base output is selected at the sixth decade counter and (b) when the input frequency is 2.9 kHz and the time base output is at the fifth decade counters.

### Question 2: (1 point)

- 2) A Basic Sweep Generator of a CRT oscilloscope has a Ramp Generator, and a Schmitt Trigger circuit. The Ramp Generator charging a capacitor  $C1 = 0.5 \mu\text{F}$  with a constant current  $I1 = 1\text{mA}$ . The Schmitt trigger circuit has its input connected to resistor  $R1 = 3 \text{K}\Omega$ , its positive FB resistor  $R2 = 9 \text{K}\Omega$ , and its bias voltage  $V_{cc} = V_{EE} = \pm 10\text{V}$ . Determine: the peak-to-peak amplitude and the period of the Sweep Generator output ramp waveform.

*Good luck*